

AISCN'26 Minor Project 2

Cyber Security Internship Report

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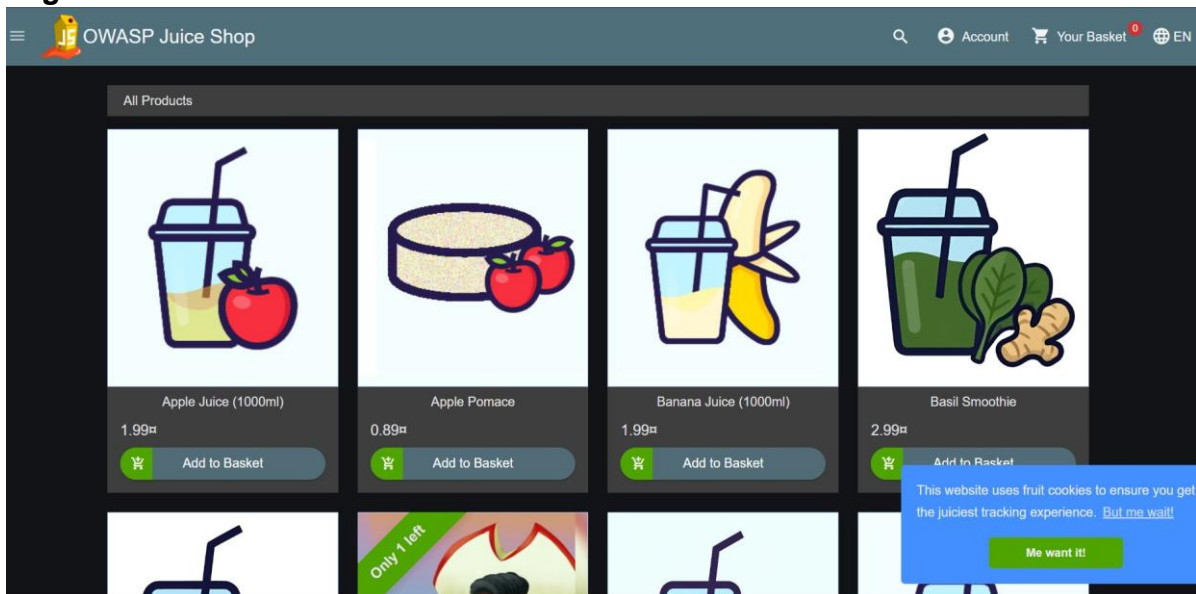
Objective

This report documents practical learning in web application reconnaissance, HTTP traffic analysis using Burp Suite, cryptographic hashing, security laboratory exercises and responsible vulnerability disclosure. The activities were carried out ethically on training platforms to understand how web applications communicate, how security analysts inspect requests and responses, and how common vulnerabilities are identified.

Task 1 – Web Application Reconnaissance

Reconnaissance was performed on the OWASP Juice Shop training application. The application interface, login page, search functionality and product pages were explored to understand publicly accessible components before any testing. During the exercise, the hosted demo later displayed a Heroku Professional Error. The issue was verified on multiple devices, indicating a server-side availability problem rather than a client-side configuration issue. This observation was recorded as part of the reconnaissance process.

Figure 1



This figure supports task 1 – web application reconnaissance. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 2

Login

Email*

Password* 

[Forgot your password?](#)

 Log in

☐ Remember me

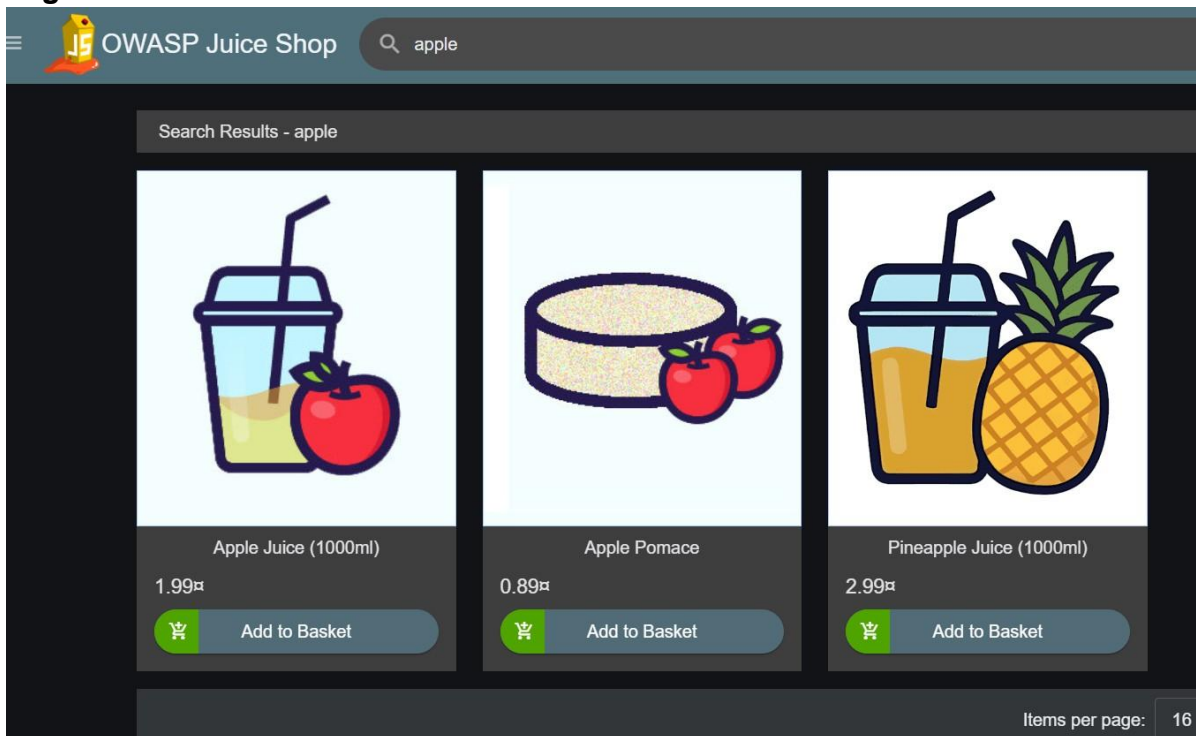
— or —

 Log in with Google

[Not yet a customer?](#)

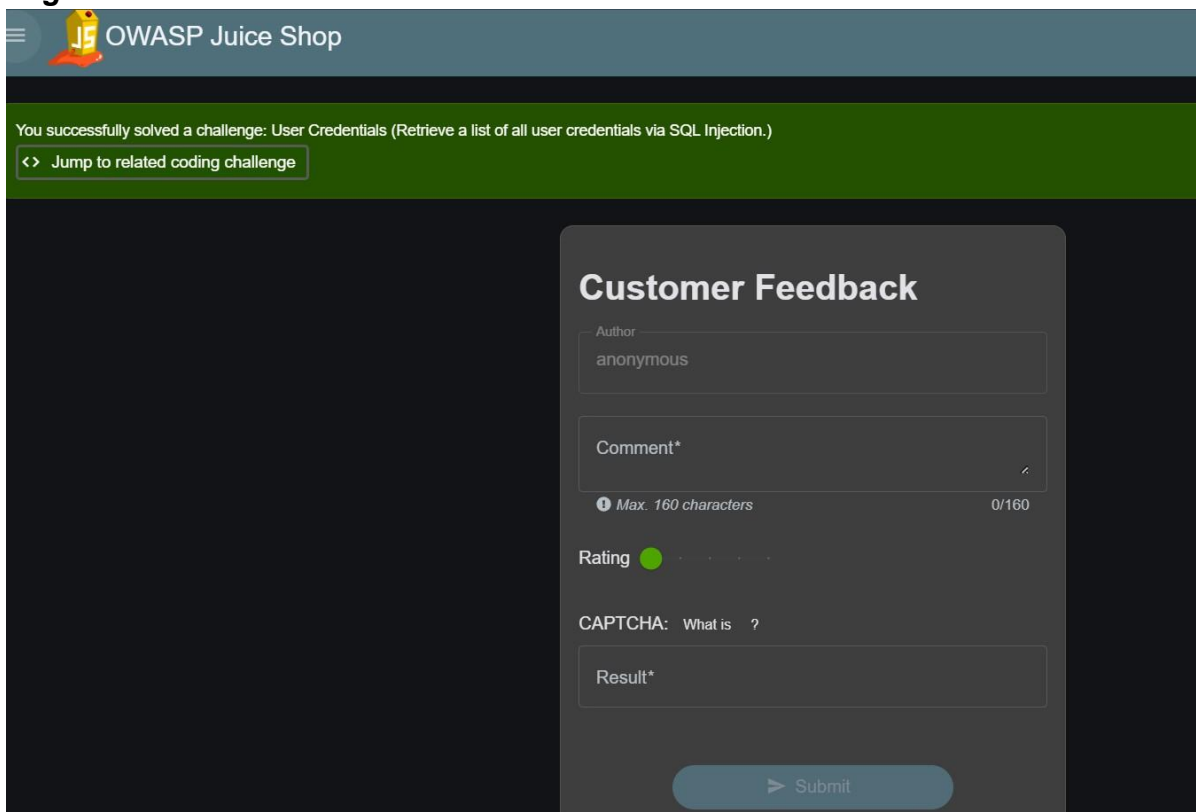
This figure supports task 1 – web application reconnaissance. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 3



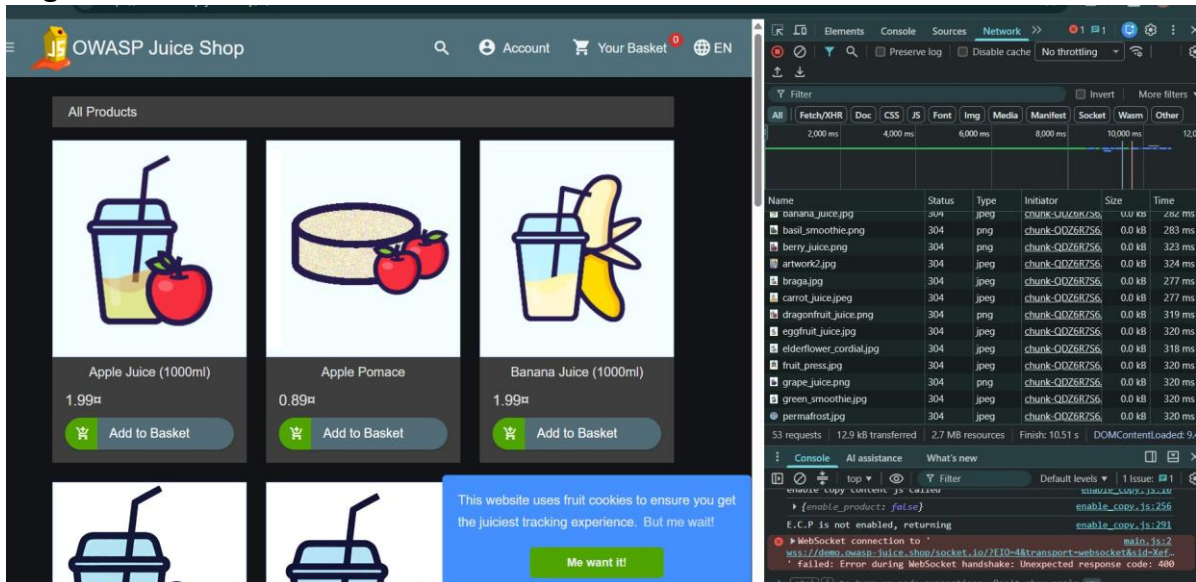
This figure supports task 1 – web application reconnaissance. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 4



This figure supports task 1 – web application reconnaissance. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 5



This figure supports task 1 – web application reconnaissance. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Task 2 – HTTP Traffic Analysis using Burp Suite

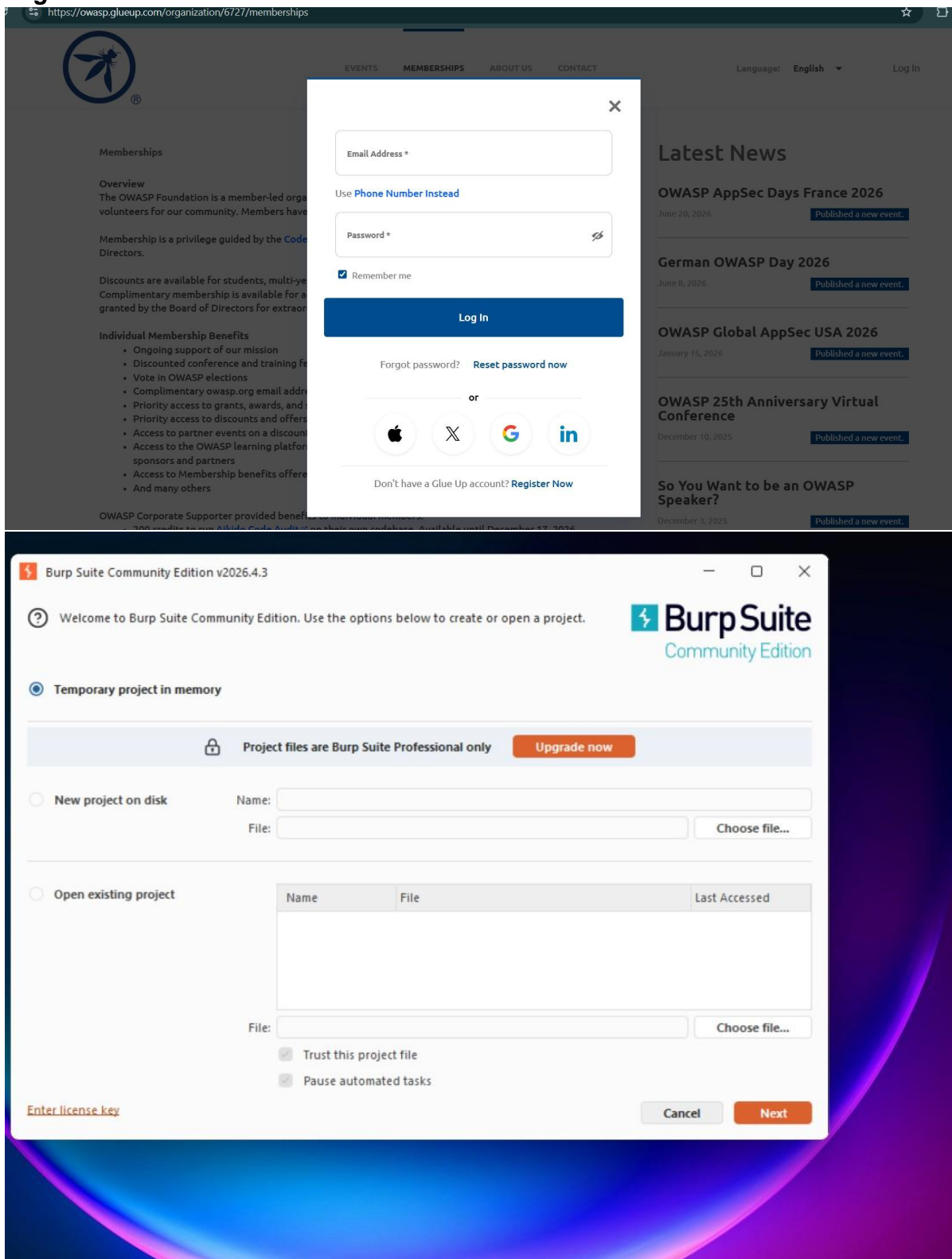
Burp Suite Community Edition was configured as an intercepting proxy. HTTP History was used to inspect browser communication. GET requests were observed while retrieving application resources, whereas POST requests were used during user input and authentication. Cookies, request headers, response headers and HTTP status codes were analyzed to understand session management and server responses.

Figure 7

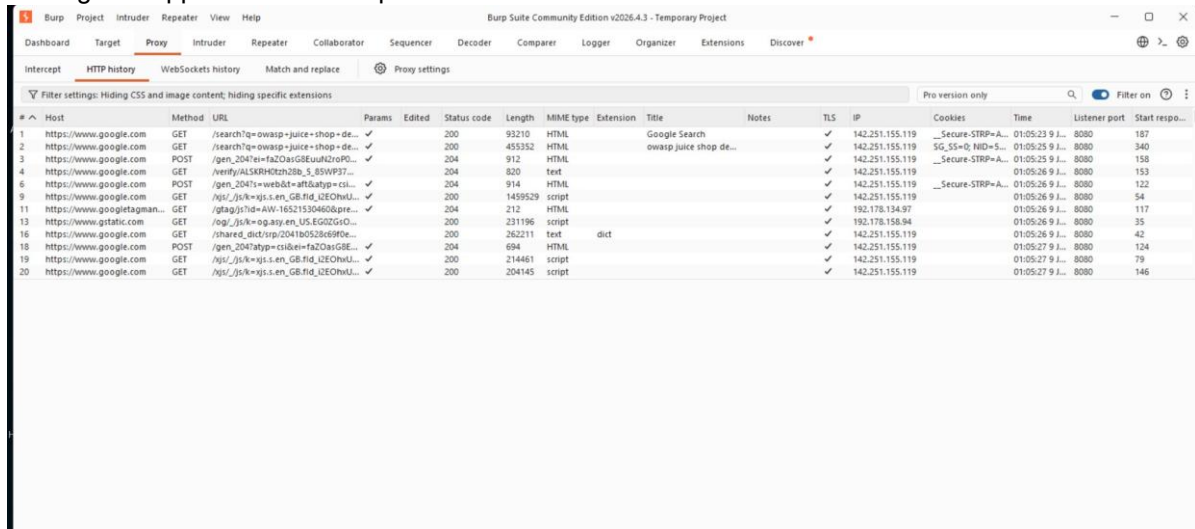
The figure consists of two main parts. The top part is a screenshot of the OWASP Juice Shop website. The header features the OWASP logo and navigation links for Projects, Chapters, Events, and About. A banner at the top encourages supporting the OWASP mission. The main content area includes a navigation bar with links like Overview, News, Challenges, Learning, CTF, Ecosystem, and Supporters. A prominent message states: "For the latest information about OWASP Juice Shop visit <https://owasp-juice.shop/>". Below this, there's a section for the "OWASP Juice Shop" project, highlighting its status as a "Flagship Project" and its classification as a "Tool" for "Builder", "Breaker", and "Defender" audiences. The bottom part of the figure is a screenshot of the Burp Suite Community Edition v2025.4.3 interface. The "Discover" tab is active, showing a "Getting started with Burp Suite" guide. This guide includes sections for "Set the target scope" (with a list of URLs to include/exclude), "Map the website" (exploring endpoints and functionality), "Analyze the attack surface" (reviewing discovered areas), and "Test for vulnerabilities" (confirming behaviors). The interface also displays a list of discovered URLs and a detailed view of a specific HTTP request (POST /login HTTP/2) with its headers and body.

This figure supports task 2 – http traffic analysis using burp suite. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 8



This figure supports task 2 – http traffic



The screenshot displays the Burp Suite interface, specifically the HTTP history tab. The interface includes a menu bar at the top with options like Burp, Project, Intruder, Repeater, View, and Help. Below the menu is a toolbar with various icons. The main area shows a list of 20 HTTP requests, each with a number, host, method, URL, status code, length, MIME type, extension, title, notes, TLS, IP, cookies, time, listener port, and start response time. The requests are filtered by 'Filter settings: Hiding CSS and image content; hiding specific extensions'. The first request is a GET to /search?q=owasp+juice+shop+de... with a status code of 200. The second request is a GET to /search?q=owasp+juice+shop+de... with a status code of 200. The third request is a POST to /gen_204?ei=faZ0asGSEuulZr0P0... with a status code of 204. The fourth request is a GET to /verify/ALSKRH0kzh28b_5_85WP37... with a status code of 204. The fifth request is a POST to /gen_204?l=web&=att0a8yp=cl... with a status code of 204. The sixth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The seventh request is a GET to /tag/js?id=AW-16521530460&pre... with a status code of 204. The eighth request is a GET to /og/_js/k=og.any.en_US.EG02GrsO... with a status code of 200. The ninth request is a GET to /shared_dict/rsp/204180528c690e... with a status code of 200. The tenth request is a POST to /gen_204?tag=cs&ei=faZ0asGSE... with a status code of 204. The eleventh request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The twelfth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The thirteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The fourteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The fifteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The sixteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The seventeenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The eighteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The nineteenth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200. The twentieth request is a GET to /js/_js/k=xjs.s.en_GB.fid_i2EOhU... with a status code of 200.

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start respo...
1	https://www.google.com	GET	/search?q=owasp+juice+shop+de...		✓	200	93210	HTML		Google Search		✓	142.251.155.119	__Secure-STRP=A...	01:05:23 9 J...	8080	187
2	https://www.google.com	GET	/search?q=owasp+juice+shop+de...		✓	200	455352	HTML		owasp juice shop de...		✓	142.251.155.119	SG_SS=0; NID=S...	01:05:25 9 J...	8080	340
3	https://www.google.com	POST	/gen_204?ei=faZ0asGSEuulZr0P0...		✓	204	912	HTML				✓	142.251.155.119	__Secure-STRP=A...	01:05:25 9 J...	8080	158
4	https://www.google.com	GET	/verify/ALSKRH0kzh28b_5_85WP37...		✓	204	820	text				✓	142.251.155.119		01:05:26 9 J...	8080	153
5	https://www.google.com	POST	/gen_204?l=web&=att0a8yp=cl...		✓	204	914	HTML				✓	142.251.155.119	__Secure-STRP=A...	01:05:26 9 J...	8080	122
6	https://www.google.com	GET	/js/_js/k=xjs.s.en_GB.fid_i2EOhU...		✓	200	1459529	script				✓	142.251.155.119		01:05:26 9 J...	8080	54
7	https://www.google.com	GET	/tag/js?id=AW-16521530460&pre...		✓	204	212	HTML				✓	192.178.134.97		01:05:26 9 J...	8080	117
8	https://www.google.com	GET	/og/_js/k=og.any.en_US.EG02GrsO...		✓	200	231196	script				✓	192.178.158.94		01:05:26 9 J...	8080	35
9	https://www.google.com	GET	/shared_dict/rsp/204180528c690e...		✓	200	262211	text	dict			✓	142.251.155.119		01:05:26 9 J...	8080	42
10	https://www.google.com	POST	/gen_204?tag=cs&ei=faZ0asGSE...		✓	204	694	HTML				✓	142.251.155.119		01:05:27 9 J...	8080	124
11	https://www.google.com	GET	/js/_js/k=xjs.s.en_GB.fid_i2EOhU...		✓	200	214461	script				✓	142.251.155.119		01:05:27 9 J...	8080	79
12	https://www.google.com	GET	/js/_js/k=xjs.s.en_GB.fid_i2EOhU...		✓	200	204145	script				✓	142.251.155.119		01:05:27 9 J...	8080	146

analysis using burp suite. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Task 3 – Cryptographic Hashing

MD5 and SHA-256 hashes of the required input were generated and compared. MD5 is fast but susceptible to collisions, making it unsuitable for modern security applications. SHA-256 provides significantly stronger collision resistance and is widely used for integrity verification and secure systems.

Figure 14

The screenshot displays the SHA256 online tool interface. On the left, a sidebar lists various cryptographic tools, with 'SHA256' selected. The main area is titled 'SHA256' and includes a description: 'This SHA256 online tool helps you calculate hashes from strings. You can input UTF-8, UTF-16, Hex, Base64, or other encodings. It also supports HMAC.' Below the title, there are 'Settings' and 'Input' sections. The 'Settings' section includes a 'Hash' button, 'Auto Update' (checked), 'Remember Input' (unchecked), 'Input Encoding' (UTF-8), 'Output Encoding' (Hex (Lower Case)), and 'Enable HMAC' (unchecked). The 'Input' section shows the input string 'AT8CR2026'. The 'Output' section displays the resulting SHA256 hash: 'b43feff928f6851dd008fdoc71b0029df4425c3d140db8959ee5fde4019c6b'. At the bottom, there is a 'Share Link' button and a browser tab labeled 'Anand - Chrome'.

This figure supports task 3 – cryptographic hashing. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 15

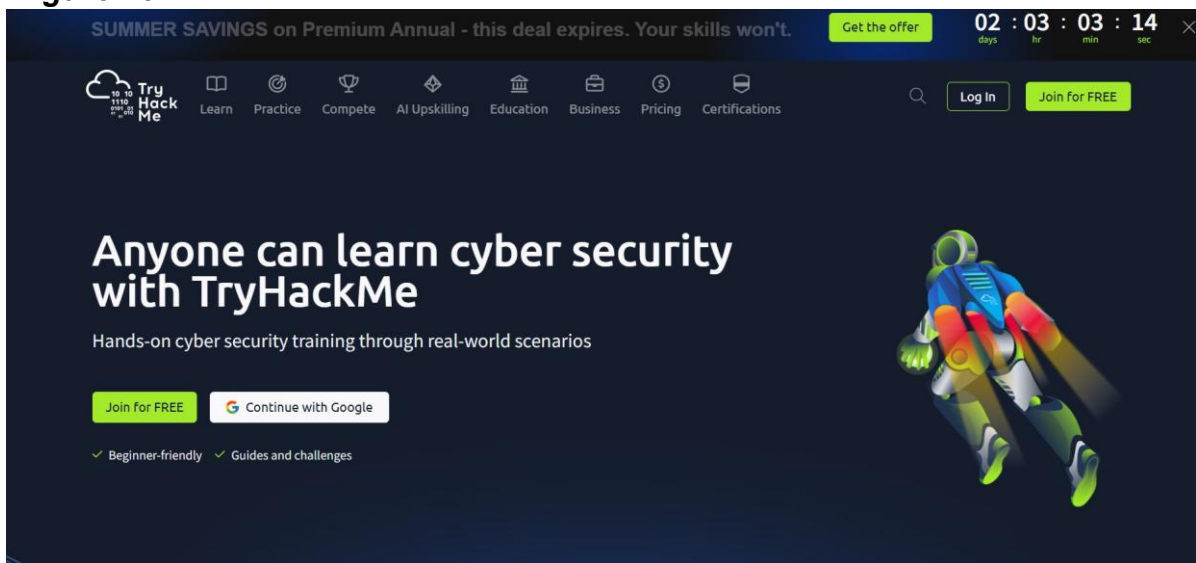
The screenshot displays the MD5 online tool interface. On the left, a sidebar lists various cryptographic tools, with 'MD5' selected. The main area is titled 'MD5' and includes a description: 'This MD5 online tool helps you calculate hashes from strings. You can input UTF-8, UTF-16, Hex, Base64, or other encodings. It also supports HMAC.' Below the title, there are 'Settings' and 'Input' sections. The 'Settings' section includes a 'Hash' button, 'Auto Update' (checked), 'Remember Input' (unchecked), 'Input Encoding' (UTF-8), 'Output Encoding' (Hex (Lower Case)), and 'Enable HMAC' (unchecked). The 'Input' section shows the input string 'AT8CR2026'. The 'Output' section displays the resulting MD5 hash: 'c92b3714e92519dd57e1efa8cce0fffb'. At the bottom, there is a 'Share Link' button and a browser tab labeled 'Anand - Chrome'.

This figure supports task 3 – cryptographic hashing. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Task 4 – Beginner Security Lab

A beginner web security lab was completed to understand the practical workflow of identifying and validating common web vulnerabilities in a controlled environment. The exercise reinforced safe testing practices and responsible learning.

Figure 16



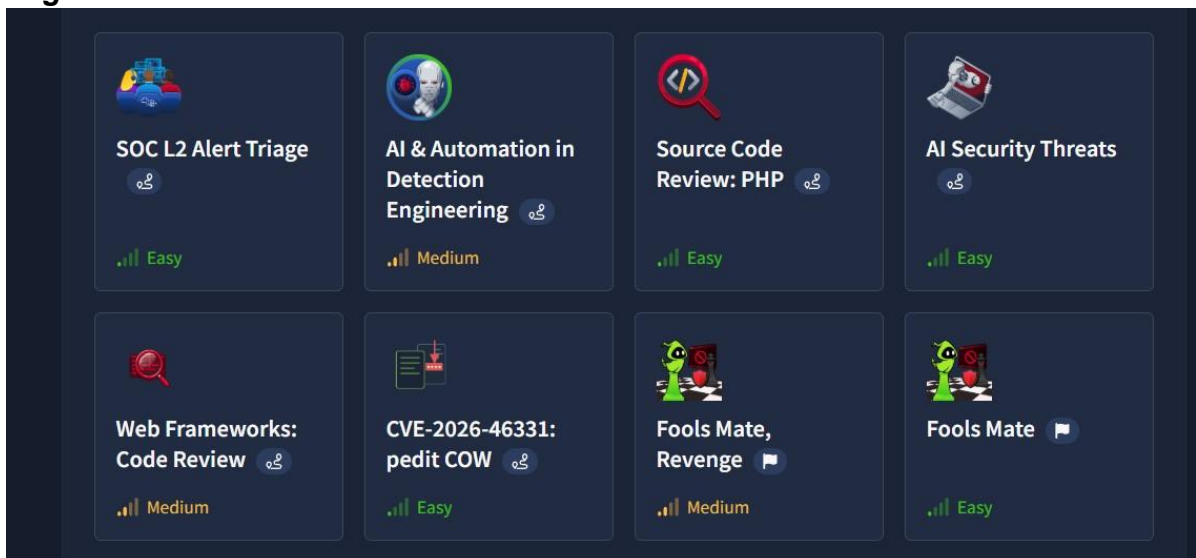
This figure supports task 4 – beginner security lab. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 17



This figure supports task 4 – beginner security lab. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 18



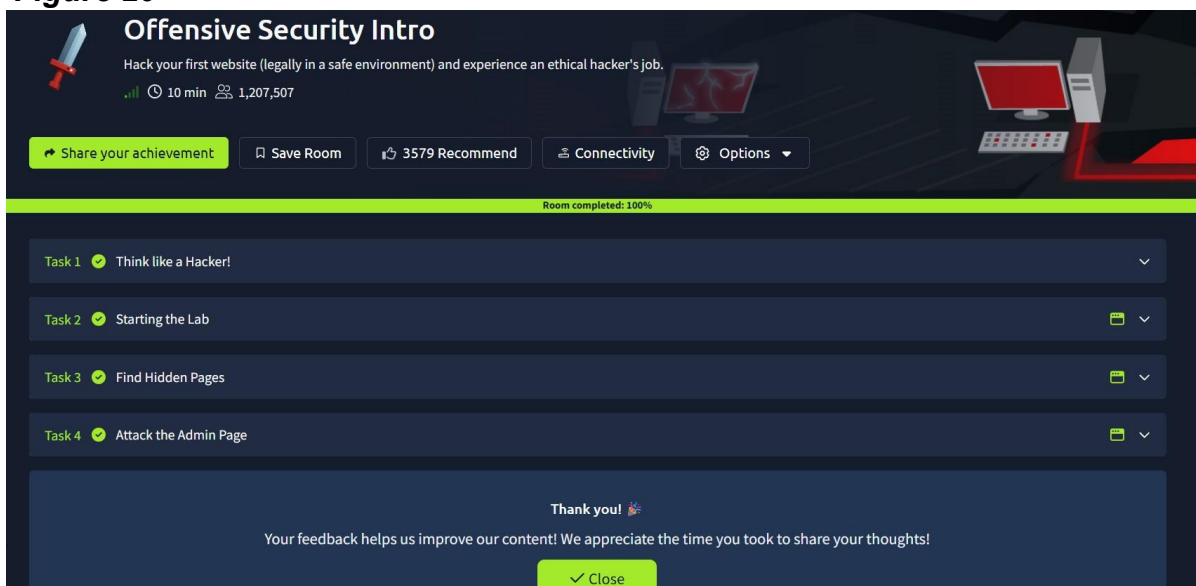
This figure supports task 4 – beginner security lab. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 19



This figure supports task 4 – beginner security lab. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Figure 20



This figure supports task 4 – beginner security lab. It records observations made during the practical exercise and forms part of the evidence collected while completing the internship assignment.

Task 5 – Bug Bounty Case Study

A public bug bounty disclosure was reviewed to understand responsible disclosure. In the selected case, a researcher identified a web application vulnerability, privately reported it through the platform, and the organization acknowledged and fixed the issue before public disclosure. The study highlighted the importance of coordinated vulnerability disclosure, secure input validation, continuous monitoring, and rewarding ethical security researchers. The biggest learning from this task is that discovering a vulnerability is only one part of cybersecurity; documenting it responsibly and helping the organization remediate it is equally important.

Conclusion

This internship project provided practical exposure to reconnaissance, traffic interception, HTTP communication, cryptographic hashing and ethical vulnerability research. Beyond completing the assigned tasks, the project emphasized careful observation, systematic analysis and responsible security practices. The experience strengthened both conceptual understanding and practical confidence in cybersecurity fundamentals.